

Inverses from Graphs and Tables

Algebra II | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or a graph, use $f^{-1}(x)$ notation, and recognize that a function and its inverse are reflections across the line $y = x$.

Vocabulary

Inverse — the relation that _____ the original.

Notation — we write _____ for the inverse of $f(x)$. Careful: this does _____ mean $1/f(x)$.

In a table — to find the inverse, _____ in every pair.

On a graph — a function and its inverse are reflections across the line _____.

Part 1 — Inverses from Ordered Pairs

Swap the coordinates in each pair.

Example 1. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{ \text{_____} \}$

Domain of $f = \text{_____}$ which becomes the _____ of f^{-1} .

You Try 1. $g = \{ (1, 5), (2, 8), (3, 11) \}$ $g^{-1} = \{ \text{_____} \}$

Part 2 — Inverses from a Table

Fill in the inverse table by trading the rows.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

Now the inverse:

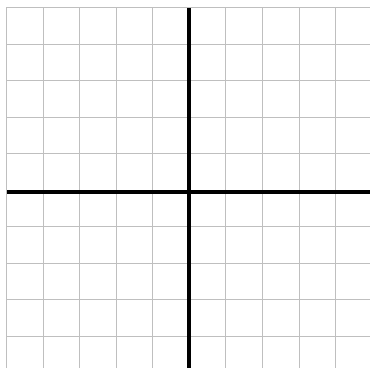
x	-6	-1	4	-7	7
$f^{-1}(x)$					

The two tables are the same information, just _____.

Part 3 — Graph a Function and Its Inverse

Each square is 1 unit. Plot f , then plot f^{-1} , then draw the line $y = x$ through the origin at 45° .

A. Plot $f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$ and its inverse.



B. What do you notice?

The two graphs are mirror images across the line _____

Which points stay put under the swap?

If (a, b) is on f , then on f^{-1} you find

$f^{-1} = \{ \text{_____} \}$

Part 4 — Domain and Range Trade Places

For any relation: domain of $f =$ _____ of f^{-1} , and range of $f =$ _____ of f^{-1} .

Function	Domain of f	Range of f	Domain of f^{-1}	Range of f^{-1}
$f(x) = x^3$	all reals	all reals	_____	_____
$f(x) = 2^x$	all reals	$(0, \infty)$	_____	_____
$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$	_____	_____

Row 2 explains Day 7: the exponential and the _____ are inverses, which is why their domains and ranges swapped.

Part 5 — Quick Check

- Does $f^{-1}(x)$ mean 1 divided by $f(x)$? _____
- If $(3, -5)$ is on the graph of f , name a point on f^{-1} . _____
- A graph and its inverse are reflections across which line? _____

Practice

Find each inverse. Show your work where there is room.

1. $p = \{ (0, 3), (1, 5), (2, 7) \}$ $p^{-1} = \{ \text{_____} \}$

2. Domain of $p =$ _____ Range of $p^{-1} =$ _____

3. A table gives f : $(2, 6), (4, 6), (5, 1)$. Write f^{-1} , then decide whether it is a function.

$f^{-1} = \{ \text{_____} \}$ A function? _____

4. That is a preview of Day 12. What went wrong in item 3?

5. If a graph contains $(-3, 7)$, name the matching point on its inverse. _____

Exit Ticket

A function h has the table below.

h : $(1, 4), (2, 9), (5, 0)$ $h^{-1} = \{ \text{_____} \}$

Domain of h^{-1} : _____ Range of h^{-1} : _____